

Event Triggered Execution

Sub-techniques (18)



Adversaries may establish persistence and/or elevate privileges using system mechanisms that trigger execution based on specific events. Various operating systems have means to monitor and subscribe to events such as logons or other user activity such as running specific applications/binaries. Cloud environments may also support various functions and services that monitor and can be invoked in response to specific cloud events.^{[1][2][3]}

Adversaries may abuse these mechanisms as a means of maintaining persistent access to a victim via repeatedly executing malicious code. After gaining access to a victim system, adversaries may create/modify event triggers to point to malicious content that will be executed whenever the event trigger is invoked.^{[4][5][6]}

Since the execution can be proxied by an account with higher permissions, such as SYSTEM or service accounts, an adversary may be able to abuse these triggered execution mechanisms to escalate their privileges.

ID: T1546

Sub-techniques: [T1546.001](#), [T1546.002](#), [T1546.003](#), [T1546.004](#), [T1546.005](#), [T1546.006](#), [T1546.007](#), [T1546.008](#), [T1546.009](#), [T1546.010](#), [T1546.011](#), [T1546.012](#), [T1546.013](#), [T1546.014](#), [T1546.015](#), [T1546.016](#), [T1546.017](#), [T1546.018](#)

- ① **Tactics:** [Privilege Escalation](#), [Persistence](#)
- ① **Platforms:** IaaS, Linux, Office Suite, SaaS, Windows, macOS

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Procedure Examples

ID	Name	Description
C0035	KV Botnet Activity	KV Botnet Activity involves managing events on victim systems via <code>libevent</code> to execute a callback function when any running process contains the following references in their path without also having a reference to <code>bioaset</code> : busybox, wget, curl, tftp, telnetd, or lua. If the <code>bioaset</code> string is not found, the related process is terminated. ^[7]
S9043	Mini Shai-Hulud	Mini Shai-Hulud has modified settings and configuration files of AI coding agents and other coding applications in order to create event triggered executions through creating hooks and runOn conditions. ^[8]
S1091	Pacu	Pacu can set up S3 bucket notifications to trigger a malicious Lambda function when a CloudFormation template is uploaded to the bucket. It can also create Lambda functions that trigger upon the creation of users, roles, and groups. ^[9]
S1164	UPSTYLE	UPSTYLE creates a <code>.pth</code> file beginning with the text <code>import</code> so that any time another process or script attempts to reference the modified item the malicious code will also run. ^[10]
S0658	XCSSET	XCSSET 's <code>dfhsebxzod</code> module searches for <code>.xcodeproj</code> directories within the user's home folder and subdirectories. For each match, it locates the corresponding <code>project.pbproj</code> file and embeds an encoded payload into a build rule, target configuration, or project setting. The payload is later executed during the build process. ^{[11][12]}

Mitigations

ID	Mitigation	Description
M1026	Privileged Account Management	Manage the creation, modification, use, and permissions associated to privileged accounts, including SYSTEM and root.
M1051	Update Software	Perform regular software updates to mitigate exploitation risk.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0010	Behavioral Detection of Event Triggered Execution Across Platforms	AN0024	Correlates unexpected modifications to WMI event filters, scheduled task triggers, or registry autorun keys with subsequent execution of non-standard binaries by SYSTEM-level processes.
		AN0025	Detects inotify or auditd configuration changes that monitor system files coupled with execution of script interpreters or binaries by cron or systemd timers.
		AN0026	Correlates launchd plist modifications with subsequent unauthorized script execution or anomalous parent-child process trees involving user agents.
		AN0027	Monitors cloud function creation triggered by specific audit log events (e.g., IAM changes, object creation), followed by anomalous behavior from new service accounts.
		AN0028	Correlates Power Automate or similar logic app workflows triggered by SaaS file uploads or email rules with data forwarding or anomalous access patterns.
		AN0029	Detects macros or VBA triggers set to execute on document open or close events, often correlating with embedded payloads or C2 traffic shortly after execution.

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